27 **C**

Since $E = \frac{n^2h^2}{8mL^2}$, we have $L = \sqrt{\frac{n^2h^2}{8mE}} \Rightarrow L \propto \frac{n}{\sqrt{mE}}$

$$\frac{L_e}{L_p} = \left(\frac{n_e}{n_p}\right)\left(\sqrt{\frac{m_pE_p}{m_eE_e}}\right)$$

$$= \left(\frac{2}{3}\right)\left(\sqrt{\frac{(1.67\times10^{-27})(0.20)}{(9.11\times10^{-31})(6.0)}}\right)$$

$$= 5.2113 \approx 5.2$$

28 **B** Diameter of an atom is $\sim 10^{-10}$ m, we approximate $\Delta x \approx 10^{-10}$ m